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Trade and Domestic Distortions: The Case of Informality

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1 Big picture

- **Question.** How does international trade interact with domestic distortions in developing countries when a large informal sector lets some firms evade taxes and labor regulations while other firms remain formal and fully exposed to them?
- **Setting.** The paper studies Brazil using a quantitative general-equilibrium model with two sectors - manufacturing (C) and services (S) - heterogeneous firms, formal and informal production, search frictions in labor markets, payroll and value-added taxes, firing costs, minimum wages, and imperfect government enforcement.
- **Core challenge.** Standard trade models usually abstract from informality or treat domestic distortions as if they affected all firms symmetrically. Here, distortions are *unequally enforced*: small and less productive informal firms escape much of the tax and regulatory burden, while larger, more productive formal firms bear it. That generates misallocation before trade reform even begins.
- **Main mechanism.** Informal firms are less distorted but also less productive, so they overproduce relative to the social optimum. Formal firms are more productive but more distorted, so they underproduce. Trade liberalization reallocates labor and output away from the first group and toward the second group.
- **Main theoretical message.** In an economy with informality, the gains from trade are not only the usual consumption gains from cheaper imports. They also include a large *reallocation effect* because trade shifts resources toward firms that were initially too small from the social point of view.
- **Main quantitative result.** For a small reduction in trade costs from $\tau_c = 2.4$ to $\tau_c = 2.3$, real income rises by about 1.32%, and the reallocation effect is larger than the mechanical effect. For a larger fall from 2.4 to 1.6, real income rises by about 24%, while the informal employment share falls only from 48.7% to 43.6%.
- **Comparative statics result.** The gains from trade are much larger when the economy starts with more informality. Under a counterfactual with no informality, the same large trade liberalization yields only about an 11% real-income gain, less than half the benchmark gain.
- **Labor-market result.** The model replicates the reduced-form evidence that informality acts as an unemployment buffer under negative labor demand shocks. But it also shows that this buffer is not a welfare buffer: informality can absorb displaced workers while worsening aggregate misallocation and real-income losses.
- **Distributional result.** Trade liberalization raises wage inequality within formal and within informal manufacturing, but once one accounts for between-sector wage gaps, overall wage inequality across workers can fall.
- **Why this paper matters.** It gives a clean quantitative framework for thinking about trade in distorted developing economies. The paper says that the effect of globalization cannot be understood without modeling informality, imperfect enforcement, and the reallocation of resources across firm types.

2 Data and measurement (key objects)

2.1 Observed data sources

The paper combines seven Brazilian datasets to discipline both the formal and informal sectors.

- **RAIS (2003–2005)**. Administrative matched employer-employee data covering the universe of formal firms and workers. It is the main source for formal-sector employment, wages, turnover, and exit.
- **PIA (2003, 2004)**. Annual manufacturing survey with revenues, capital, investment, intermediate inputs, and employment.
- **PAS (2003, 2004)**. Annual services survey with revenues, capital, investment, intermediate inputs, and employment.
- **PAC (2003, 2004)**. Annual retail and commerce survey, used together with PAS for the broader service sector.
- **SECEX (2003)**. Customs records with firm-level exports and imports.
- **ECINF (2003)**. Urban informal-economy survey, representative of firms with up to five employees, with matched information on businesses and workers and direct measures of firms' and workers' formal status.
- **PME (2003)**. Rotating household panel covering the six major metropolitan areas, used for employment shares and labor-market transition rates.

2.2 Sample restrictions and key definitions

- The paper restricts the analysis to the six metropolitan areas covered by PME so that all data sources line up geographically.
- It excludes agriculture, mining, coal, and oil and gas, mainly because the informal-firm data are urban and do not cover those sectors well.
- It excludes the public sector.
- A worker is **informal** if she does not hold a formal labor contract. Self-employed workers are also treated as informal.
- A firm is **informal** if it is not registered with the tax authorities; in Brazil this means not having a CNPJ tax identification number.
- Because ECINF is available only in 2003, the baseline cross-sectional moments are built around 2003, with 2004 used when turnover or transitions require dynamics.

2.3 Five empirical facts that motivate the model

- **Fact 1: informality is huge and unemployment flows disproportionately into informality.** Using PME, the paper shows that 48.2% of workers are informally employed. The

employment shares are

$$h_{Ci} = 0.059, \quad h_{Cf} = 0.106, \quad h_{Si} = 0.351, \quad h_{Sf} = 0.334,$$

with unemployment equal to 0.151. The transition rate from unemployment to informal employment is 45.7%, versus only 26.4% to formal employment.

- **Fact 2: informality falls sharply with firm size.** In ECINF, regressing an informal-status indicator on firm employment gives slopes of about -0.212 in manufacturing and -0.216 in services. The model therefore needs a size-dependent cost of staying informal.
- **Fact 3: informal firms are much less productive on average.** Conditional on firm size, formal firms have much higher revenue per worker than informal firms. In manufacturing, the formal-informal gap is about 1.76 log points for firms of size one and still about 1.03 log points for firms of size five. In services, the corresponding gaps are about 1.23 and 0.38 log points.
- **Fact 4: informal workers are paid less on average.** Conditional on firm size, the formal wage premium is about 0.59 log points in manufacturing for size-one firms and 0.23 for size-five firms. In services, the premium starts at about 0.19 for size-one firms; by size five the within-size premium can reverse, but informal firms remain lower paying on average because they are so much smaller.
- **Fact 5: turnover falls with size, but exporters have higher turnover conditional on size.** In RAIS, firm-level turnover is strongly decreasing in employment size, while exporters exhibit higher turnover conditional on size. This matters because trade openness reallocates activity toward both larger firms and exporters, which pulls turnover in opposite directions.

2.4 Unobservables and timing

- The central firm-level unobservable is idiosyncratic productivity z , which evolves stochastically over time.
- Firms also differ in formal versus informal status and in whether they export.
- Labor market objects such as the value of unemployment J^u , vacancy-filling rates, and bargained wages are endogenous equilibrium objects.
- Within a period, incumbent firms start with current productivity z and workforce ℓ , decide whether to remain active, formalize, or exit, then adjust employment, bargain over wages, produce, and only afterward draw next period's productivity z' .

2.5 Empirical applications

Once estimated, the model is used for four main exercises:

- small trade-cost shocks and their decomposition into mechanical and reallocation effects,
- larger trade reforms,
- trade reforms under stricter enforcement / less informality,
- negative labor-demand shocks and their effect on informality, unemployment, productivity, and wage inequality.

3 Models and empirical specifications (all equations + interpretation)

3.1 Consumers and final demand

The representative household has linear period utility in the final good and discounts at rate r :

$$U = \sum_{t=1}^{\infty} \frac{Y_t}{(1+r)^t}. \quad (1)$$

- **Interpretation.** This is a static welfare criterion inside a steady-state environment. The final good is the numeraire consumption object whose price is $P_{Y,t}$.

The final good is assembled competitively from manufacturing and services composite goods:

$$Y_t = \hat{C}_t^\zeta \hat{S}_t^{1-\zeta}. \quad (2)$$

- **Interpretation.** Consumers spend a share ζ on the manufacturing composite and $1 - \zeta$ on the services composite.
- **Why it matters.** Trade shocks operate directly through the manufacturing side, but because services enter final demand and production, the service sector also responds in equilibrium.

3.2 Composite goods and intersectoral linkages

Sector-specific composite goods are CES bundles of differentiated varieties:

$$\mathcal{C}_t = \left(\int_0^{N_{Ct}} q_{Ct}(n)^{\frac{\sigma_C-1}{\sigma_C}} dn \right)^{\frac{\sigma_C}{\sigma_C-1}}, \quad \mathcal{S}_t = \left(\int_0^{N_{St}} q_{St}(n)^{\frac{\sigma_S-1}{\sigma_S}} dn \right)^{\frac{\sigma_S}{\sigma_S-1}}. \quad (3)$$

Sector-specific intermediate-input bundles are Cobb-Douglas composites of the two sectoral composites:

$$\iota_k = \tilde{C}^{\lambda_k} \tilde{S}^{1-\lambda_k}, \quad k \in \{C, S\}. \quad (4)$$

- **Interpretation.** Both sectors use manufacturing and service composites as inputs.
- **Key point.** This roundabout production structure is one reason trade shocks propagate beyond directly traded manufacturing goods.

3.3 Firm technology and productivity dynamics

A firm in sector $k \in \{C, S\}$, with productivity z , employment ℓ , and intermediate input usage ι_k , produces

$$q_k(z, \ell, \iota_k) = z \ell^{\delta_k} \iota_k^{1-\delta_k}. \quad (5)$$

Productivity evolves as an AR(1):

$$\ln z' = \rho_k \ln z + \sigma_k^z \varepsilon, \quad \rho_k \in (0, 1), \quad \varepsilon \sim N(0, 1). \quad (6)$$

Under monopolistic competition, gross revenue as a function of output takes the form

$$R_k(q) = \left(\frac{X_k}{P_k^{1-\sigma_k}} \right)^{1/\sigma_k} q^{(\sigma_k-1)/\sigma_k}, \quad (7)$$

where X_k is total expenditure on the sector- k composite good and P_k is the corresponding CES price index.

Because intermediates are freely adjusted, firm-level value added is proportional to gross revenue:

$$VA_k(z, \ell) = \frac{\sigma_k - (1 - \delta_k)(\sigma_k - 1)}{\sigma_k} R_k(z, \ell). \quad (8)$$

- **Interpretation.** More productive firms produce more and earn more revenue, but markups depend on the sectoral elasticity of substitution σ_k .
- **Why this matters.** Heterogeneous productivity and endogenous firm size are what allow formal and informal firms to sort differently and create misallocation.

3.4 Hiring costs and dynamic labor adjustment

Hiring is costly. The hiring cost function is

$$C_k^h(\ell, v) = \frac{h_k}{\gamma_{k1}} \left(\frac{v}{\ell^{\gamma_{k2}}} \right)^{\gamma_{k1}}, \quad (9)$$

where v is the number of posted vacancies. If a firm expands from ℓ to ℓ' , the vacancy requirement is $v = (\ell' - \ell)/\mu_v$, so the cost of expansion can be written

$$H_{kj}(\ell, \ell') = (\mu_v)^{-\gamma_{k1}} \frac{h_k}{\gamma_{k1}} \left(\frac{\ell' - \ell}{\ell^{\gamma_{k2}}} \right)^{\gamma_{k1}}, \quad j \in \{f, i\}. \quad (10)$$

Formal firms also face firing costs:

$$F(\ell, \ell') = \kappa(\ell - \ell'). \quad (11)$$

- **Interpretation.** Hiring is convex and can exhibit scale economies. Firing costs apply only to formal firms.
- **Why this matters.** These two ingredients generate turnover patterns, unemployment, and wage dispersion, all of which are crucial to the paper's labor-market results.

3.5 Formal firms

Formal firms pay value-added and payroll taxes, comply with the minimum wage, and pay firing costs. Their profit function is

$$\pi_{kf}(z, \ell, \ell') = (1 - \tau_y)VA_k(z, \ell') - C_{kf}(z, \ell, \ell') - c_k, \quad (12)$$

with total labor-adjustment cost

$$C_{kf}(z, \ell, \ell') = \begin{cases} (1 + \tau_w) \max\{w_{kf}(z, \ell'), \underline{w}\} \ell' + H_{kf}(\ell, \ell') & \text{if } \ell' > \ell, \\ (1 + \tau_w) \max\{w_{kf}(z, \ell'), \underline{w}\} \ell' + \kappa(\ell - \ell') & \text{if } \ell' \leq \ell. \end{cases} \quad (13)$$

The formal firm's value function is

$$V_{kf}(z, \ell) = (1 - \alpha_k) \max \left\{ 0, \max_{\ell'} \left[\pi_{kf}(z, \ell, \ell') + \frac{1}{1+r} \mathbb{E}_{z'|z} V_{kf}(z', \ell') \right] \right\}. \quad (14)$$

- **Interpretation.** Formal firms choose whether to continue, how many workers to employ next period, and whether to pay the fixed cost of staying active.
- **Distortion.** Taxes, minimum wages, and firing costs all compress formal profits and push the formal sector below the efficient scale.

3.6 Informal firms and the cost of informality

Informal firms do not pay formal-sector taxes and labor regulations, but they do face an expected cost of being informal:

$$\pi_{ki}(z, \ell, \ell') = V A_k(z, \ell') - K^{\text{inf}}(z, \ell') - C_{ki}(z, \ell, \ell') - c_k, \quad (15)$$

with

$$p_{ki}(\ell') = \tilde{a}_k \exp\{\tilde{b}_k(\ell' - 1)\}, \quad K^{\text{inf}}(z, \ell') = p_{ki}(\ell') R_k(z, \ell'). \quad (16)$$

Their labor cost is

$$C_{ki}(z, \ell, \ell') = \begin{cases} w_{ki}(z, \ell') \ell' + H_{ki}(\ell, \ell') & \text{if } \ell' > \ell, \\ w_{ki}(z, \ell') \ell' & \text{if } \ell' \leq \ell. \end{cases} \quad (17)$$

The informal firm's value function is

$$V_{ki}(z, \ell) = (1 - \alpha_k) \max \left\{ 0, \max_{\ell'} \left[\pi_{ki}(z, \ell, \ell') + \frac{1}{1+r} \mathbb{E}_{z'|z} V_{ki}(z', \ell') \right], \max_{\ell'} \left[\pi_{kf}(z, \ell, \ell') + \frac{1}{1+r} \mathbb{E}_{z'|z} V_{kf}(z', \ell') \right] \right\}. \quad (18)$$

- **Interpretation.** Informal firms can stay informal, exit, or formalize. Formal firms, by assumption, do not move back into informality.
- **Key economic logic.** The size-dependent penalty $p_{ki}(\ell')$ makes informality attractive for small firms but increasingly costly for larger ones. That is how the model matches the sharp empirical relationship between size and informality.
- **Important equilibrium implication.** Because formal firms cannot revert to informality, the productivity distributions of formal and informal firms overlap in equilibrium. This mirrors the data.

3.7 Entry

Potential entrants pay a sunk entry cost c_k^e , draw productivity from the ergodic distribution, and then choose to enter as formal, informal, or not at all:

$$V_{kj}^e(z) = \max_{\ell'} \left[\pi_{kj}(z, 1, \ell') + \frac{1}{1+r} \mathbb{E}_{z'|z} V_{kj}(z', \ell') \right], \quad j \in \{f, i\}, \quad (19)$$

$$V_k^e = E_z \max\{V_{ki}^e(z), V_{kf}^e(z), 0\}, \quad (20)$$

with free entry when

$$V_k^e = c_k^e. \quad (21)$$

- **Interpretation.** Entry thresholds into informality and formality depend on productivity, labor costs, and future export opportunities.
- **Trade channel.** Lower trade costs increase the value of entering formally in manufacturing because exporters are formal and export prospects improve.

3.8 Search, matching, and unemployment

The labor market has random search and matching frictions. Total matches are

$$m(v_{Cf}, v_{Ci}, v_{Sf}, v_{Si}, L^u) = \phi \tilde{v}^\xi (L^u)^{1-\xi}, \quad (22)$$

where $\tilde{v} = v_{Cf} + v_{Ci} + v_{Sf} + v_{Si}$.

The vacancy-filling rate is

$$\mu_v = \phi \left(\frac{L^u}{\tilde{v}} \right)^{1-\xi}, \quad (23)$$

and the job-finding probability into sector k and type j is

$$\mu_{kj}^e = \frac{m_{kj}}{L^u} = \left(\frac{v_{kj}}{\tilde{v}} \right) \phi \left(\frac{\tilde{v}}{L^u} \right)^\xi. \quad (24)$$

- **Interpretation.** Formal and informal firms compete for the same pool of unemployed workers.
- **Why it matters.** Trade shocks affect unemployment both directly, through turnover and vacancy posting, and indirectly, because unemployment flows disproportionately into informal jobs.

3.9 Wages and bargaining

Wages are set by Nash bargaining between firms and workers after hiring decisions are made. For formal firms, firm and worker surpluses are

$$S_{kf}^e(z, \ell') = (1 - \tau_y) V A_k(z, \ell') - (1 + \tau_w) w_{kf}(z, \ell') \ell' + \frac{1}{1+r} E_{z'|z} V_{kf}(z', \ell'), \quad (25)$$

$$S_{kf}^u(z, \ell') = w_{kf}(z, \ell') + \frac{1}{1+r} J_{kf}^e(z, \ell') - \left[b_0 P_Y + b^u + \frac{1}{1+r} J^u \right]. \quad (26)$$

The Nash bargaining rule is

$$S_{kf}^u(z, \ell') = \beta [S_{kf}^e(z, \ell') + S_{kf}^u(z, \ell')], \quad (27)$$

which implies the wage schedule

$$(1 + \beta \tau_w) w_{kf}(z, \ell') = (1 - \beta) \left[b_0 P_Y + b^u + \frac{1}{1+r} (J^u - J_{kf}^e(z, \ell')) \right] + \beta \left[(1 - \tau_y) \frac{V A_k(z, \ell')}{\ell'} + \frac{1}{1+r} \mathbb{E}_{z'|z} \frac{V_{kf}(z', \ell')}{\ell'} \right]. \quad (28)$$

Actual formal wages are

$$w_{kf}(z, \ell') = \max\{w_{kf}^u(z, \ell'), w_{kf}^{res}(z, \ell')\}, \quad \text{and firms pay } \max\{w_{kf}(z, \ell'), \underline{w}\}. \quad (29)$$

Informal wages are analogous but exclude payroll taxes, unemployment insurance, and the legal minimum wage:

$$w_{ki}(z, \ell') = \max\{w_{ki}^u(z, \ell'), w_{ki}^{res}(z, \ell')\}. \quad (30)$$

- **Interpretation.** Wages depend on productivity, firm size, and the worker's outside option.
- **Why wage dispersion matters.** The model needs equilibrium wage dispersion to match the fact that informal firms pay lower wages and to make the minimum wage relevant quantitatively.

3.10 Open economy, importing, and exporting

Only manufacturing varieties are internationally traded. The price index of the domestic manufacturing composite combines home and foreign varieties, with imports paying ad valorem tariffs $(\tau_a - 1)$ and iceberg trade costs τ_c . Domestic demand for home varieties is standard CES demand, and foreign demand for exporters is

$$Q_{F,C}(n) = D_F^*(p_x^*(n))^{-\sigma_C}.$$

Export revenues imply that, conditional on exporting, all exporters ship the same fraction η_o of output abroad. The export share of gross revenue is

$$\frac{R_C^x(z, \ell') - R_C^{\text{dom},x}(z, \ell')}{R_C^x(z, \ell')} = 1 - e^{-\sigma_C d_F}. \quad (33)$$

Exporter value added is a multiplicative “magnification” of domestic value added:

$$VA_C^x(z, \ell') = \exp\left(\frac{\sigma_C}{\sigma_C - 1} \Lambda_C d_F\right) VA_C^d(z, \ell'). \quad (34)$$

Hence the export decision is

$$\mathbb{I}_C^x(z, \ell') = \begin{cases} 1 & \text{if } VA_C^x(z, \ell') - f_x > VA_C^d(z, \ell'), \\ 0 & \text{otherwise.} \end{cases} \quad (35)$$

- **Interpretation.** Lower trade costs increase d_F , raise the profitability of exporters, and induce more manufacturing firms to export.
- **Key implication.** Because exporters are formal, trade reform changes not just export participation but also the relative value of formality.

3.11 Equilibrium

The steady-state equilibrium has the following ingredients:

- firms choose entry, exit, formality, employment, and export status optimally,
- wage schedules solve the bargaining problem,
- labor markets clear,

- product markets clear,
- trade is balanced,
- the government budget balances,
- aggregate income equals total wages and profits plus tariff revenues minus entry, hiring, operating, and export costs,
- the distributions of firms over (z, ℓ) , by sector and formality status, are stationary.
- **Interpretation.** The model is solved in steady state, but it still has rich micro dynamics because firms move across states, adjust labor, exit, enter, and formalize over time.

3.12 How trade affects informality and unemployment in the model

The paper emphasizes two broad sets of mechanisms.

- **Melitz-type selection and export mechanisms.** Lower τ_a or τ_c raises export opportunities in manufacturing, increases exporters' profitability and wages, and pushes up the value of formal status for productive informal firms that may later export. This tends to reduce informality.
- **Input-cost and demand-propagation mechanisms.** Cheaper imported manufacturing inputs lower costs through the roundabout production structure and raise aggregate productivity, which changes entry, exit, and formalization decisions in both sectors.
- **Unemployment channel.** Trade openness reallocates activity toward larger firms, which tends to reduce turnover, but also toward exporters, whose revenues become more sensitive to shocks, which tends to increase turnover. The net unemployment effect is therefore ambiguous a priori.

Quantitatively, in the estimated model, trade liberalization shrinks informality and raises unemployment slightly in the benchmark equilibrium because the exporter-sensitivity force dominates the stability-from-size force.

3.13 Quantitative decompositions of trade gains

For a small change in trade costs, the paper uses two decompositions.

First,

$$\Delta \log \mathcal{Y} \approx \Delta \log \mathcal{Y}^{ME} + \Delta \text{Reallocation Effect}, \quad (37)$$

where $\Delta \log \mathcal{Y}^{ME}$ is the *mechanical effect*: the direct cost reduction from cheaper imports, propagated through the input-output structure while holding allocations fixed.

Second,

$$\Delta \log \mathcal{Y} \approx \Delta \log \mathcal{Y}^{DF} + \Delta \text{Reallocation Effect}. \quad (39)$$

Here $\Delta \log \mathcal{Y}^{DF}$ is the gain from the same trade-cost shock in a hypothetical undistorted economy with the same production network and import share.

- **Interpretation.** These decompositions separate ordinary trade gains from extra gains caused by reallocation across distorted firms.

- **Why this is central.** The paper’s main quantitative message is precisely that this reallocation term is large when informality is pervasive.

4 Identification and instruments (what makes the estimates credible)

4.1 What makes this paper different

This paper does not rely on a traditional reduced-form instrument. Its empirical strategy is structural: map a rich dynamic model into moments from many datasets, estimate the model by indirect inference, and then use the fitted model for counterfactuals.

- **Key idea.** Credibility comes from how distinct moments pin down distinct parameters and from whether the estimated model reproduces the five facts that motivated the model in the first place.
- **Why this matters.** The paper is about aggregate counterfactuals - not local reduced-form treatment effects - so it needs a fully specified equilibrium structure.

4.2 Fixed parameters from institutions and external sources

Some parameters are pinned down outside the indirect-inference step:

- $\tau_c = 2.4$ from the Head-Ries index and trade/input-output data.
- $\zeta = 0.296$, $\lambda_C = 0.645$, and $\lambda_S = 0.291$ from Brazilian national accounts.
- Interest rate $r = 0.08$.
- Value-added tax $\tau_y = 0.293$, payroll tax $\tau_w = 0.375$, and import tariff $\tau_a - 1 = 0.12$.
- Firing cost $\kappa = 1956.7$ reais, minimum wage $\underline{w} = 2880$, and unemployment benefit $b^u = 1644$.
- Matching elasticity $\xi = 0.5$ and worker bargaining power $\beta = 0.5$.
- Matching efficiency $\phi = 0.576$ is set so the model matches the unemployment-to-employment transition rate in the data.
- **Interpretation.** The authors map Brazil’s institutional environment into model objects rather than leaving those objects free to absorb everything.

4.3 Indirect inference estimator

The paper estimates 27 remaining parameters using 74 moments and auxiliary-model coefficients. The objective function is

$$\mathcal{L}(\Omega) = \sum_i W_i |m_i^{\text{Model}}(\Omega) - m_i^{\text{Data}}|.$$

The estimation treats the endogenous equilibrium objects Ψ_C , Ψ_S , and ϑ^{J^u} as parameters to be solved along with the structural parameters. After minimization, the flow value of unemployment b_0 and the foreign-demand shifter D_F^* are backed out from equilibrium relationships.

- **Interpretation.** Rather than estimate every equilibrium object directly, the procedure nests equilibrium inside the minimum-distance problem.
- **Practical point.** The weights W_i are chosen to prioritize the most economically important facts, especially the five stylized facts and the unemployment rate.

4.4 What identifies what

The paper provides heuristic identification arguments linking moments to parameters.

- **Hiring-cost level $\tilde{h}_k$.** Identified from turnover rates and the unemployment rate.
- **Hiring-cost scale economies γ_{k2} .** Identified from the relationship between turnover and firm size.
- **Hiring-cost convexity γ_{k1} .** Identified from wage dispersion across firms of different size and export status.
- **Fixed operating costs c_k .** Identified from how exit probabilities vary with size.
- **Exogenous destruction rates α_k .** Identified from average exit rates.
- **Elasticities of substitution σ_k .** Identified from the relationship between revenues and employment size.
- **Productivity persistence ρ_k .** Identified from serial correlation in firm employment and firm revenues.
- **Shock volatility σ_k^z .** Identified from the cross-sectional dispersion of firm employment and revenues.
- **Fixed export cost f_x .** Identified from the exporter share in manufacturing.
- **Informality-cost intercept and convexity $(\tilde{a}_k, \tilde{b}_k)$.** Identified from the share of informal employment, the informal-firm size distribution, and the fraction of firms that are informal conditional on employment size.

4.5 Estimated parameter values and interpretation

The main estimates are reported in Table VII.

- Informality costs are more steeply increasing with size in manufacturing than in services: $(\tilde{a}_C, \tilde{b}_C) = (0.205, 0.071)$ versus $(\tilde{a}_S, \tilde{b}_S) = (0.379, 0.010)$.
- Hiring is more convex in services: $\gamma_{C1} = 1.665$ versus $\gamma_{S1} = 5.791$.
- Elasticities of substitution are about $\sigma_C = 5.023$ and $\sigma_S = 2.955$.
- Productivity is highly persistent in both sectors: $\rho_C = 0.977$, $\rho_S = 0.963$.
- Productivity shocks are more volatile in services: $\sigma_C^z = 0.210$, $\sigma_S^z = 0.409$.
- Labor's share in production is much larger in services: $\delta_C = 0.256$, $\delta_S = 0.516$.

- The fixed cost of exporting is large: $f_x \approx 54,989$ reais.
- The estimated flow utility of unemployment is strongly negative: $b_0 = -0.801$ times real income per capita in the model.

4.6 Model fit and credibility

The model fits the key moments well.

- It reproduces the large overall informality share and the much larger unemployment-to-informality flow than unemployment-to-formality flow.
- It matches the steep decline in informality with firm size.
- It reproduces the fact that formal firms are more productive and pay higher wages on average.
- It matches the relationship between turnover, firm size, and export status.
- It also replicates wage-size and exporter premia, which is important for the wage-inequality counterfactuals later in the paper.

The main shortcomings are that the model underestimates the dispersion of firm size and revenues in the formal sector and somewhat overstates revenues among informal manufacturing firms. But the broad cross-sectional structure of the data is well captured.

4.7 Relation to the literature

- Relative to the *trade and informality* reduced-form literature, the contribution is to study aggregate general-equilibrium effects rather than relative local labor-market effects.
- Relative to *Melitz* models, the paper adds imperfect enforcement, formality choice, unemployment, wage bargaining, minimum wages, firing costs, and intersectoral linkages.
- Relative to the *misallocation* literature, the paper's novelty is that distortions arise endogenously from imperfect enforcement and are tied to informality.
- Relative to the *labor-search* literature, the paper uses unemployment and turnover dynamics to connect trade shocks to formality transitions and wage dispersion.

5 Tables and figures

5.1 Table I: Summary of datasets

Read:

- The paper's empirical design is built on unusually rich data: administrative data for the formal sector, special surveys for the informal sector, firm surveys for inputs and revenues, customs data for exports and imports, and worker panel data for labor-market transitions.
- RAIS is what gives the authors the universe of formal firms and workers. ECINF is what makes it possible to see small informal firms directly rather than inferring them residually.

- PME is crucial because it measures employment-status transitions and therefore disciplines the unemployment and informality margins jointly.
- This table makes clear that the model is not calibrated to a single dataset; it is disciplined by several data sources that each pin down a different margin.

TABLE I
SUMMARY OF DATASETS.

Dataset	Source	Description
Relação Anual de Informações Sociais <i>RAIS</i> Years: 2003–2005	Ministry of Labor	Administrative matched employer-employee dataset. Covers all formal firms and workers. Detailed information on firms and workers, but no information on firm-level revenues, capital and expenditures with intermediate inputs.
Pesquisa Industrial Anual <i>PIA</i> Years: 2003, 2004	IBGE	Survey data on Manufacturing firms. Firm-level information such as revenues, capital, investment, expenditures with intermediate inputs, employment. Covers all firms with 30 employees or more; random sample of smaller firms.
Pesquisa Anual dos Serviços <i>PAS</i> Years: 2003, 2004	IBGE	Survey data on Service-sector firms. Firm-level information such as revenues, capital, investment, expenditures with intermediate inputs, employment. Covers all firms with 20 employees or more; random sample of smaller firms.
Pesquisa Anual do Comércio <i>PAC</i> Years: 2003, 2004	IBGE	Survey data on Retail and Commerce firms. Firm-level information such as revenues, capital, investment, expenditures with intermediate inputs, employment. Covers all firms with 20 employees or more; random sample of smaller firms.
Secretaria de Comércio Exterior <i>SECEX</i> Year: 2003	Ministry of Industry, Foreign Trade and Services	Administrative customs data. Export and import values at the firm level.
Economia Informal Urbana <i>ECINF</i> Year: 2003	IBGE	Survey data. Matched employer-employee dataset with detailed information on formal and informal firms and their workers. Representative sample of small businesses (firms with 5 employees or less). Information on formal status of the firm and its workers.
Pesquisa Mensal de Emprego <i>PME</i> Year: 2003	IBGE	Survey data. Rotating panel of households that covers the 6 main metropolitan areas in Brazil.

TABLE II
EMPLOYMENT SHARES AND TRANSITION RATES.

	Share of Workers	Transition Rates From Unemp.
Informal Manufacturing (Ci)	0.059	0.058
Formal Manufacturing (Cf)	0.106	0.059
Informal Services (Si)	0.351	0.399
Formal Services (Sf)	0.334	0.206
Unemployment	0.151	0.279
Share of Informal Employment		0.482
Transition Rate from Unemp. to Informal Employment		0.457
to Formal Employment		0.264

Note: Data source: 2003 PME.

5.2 Table II: Employment shares and transition rates

Read:

- Informality is quantitatively central: informal manufacturing accounts for 5.9% of the workforce and informal services for 35.1%, so informal employment totals 48.2% of all workers.
- Unemployment is also sizable at 15.1%.
- The key labor-market asymmetry is in the flows out of unemployment: the transition rate to informality is 45.7%, versus only 26.4% to formality.
- This is the reason the model treats unemployment and informality jointly: any shock that changes unemployment mechanically changes the informal sector too.

5.3 Table III and Table IV: Size, productivity, and wages in formal and informal firms

Read:

- Table III shows a very steep size gradient in formality: the informal-status coefficient on employment is around -0.21 in both sectors.
- Table IV shows that formal firms are much more productive conditional on size. In manufacturing the formal-informal revenue-per-worker gap is about 1.76 log points at size one and 1.03 at size five; in services it is about 1.23 and 0.38.

TABLE III
FIRM-LEVEL INFORMALITY STATUS VS. FIRM-LEVEL EMPLOYMENT.

	Dep. Variable: <i>Informal Status Indicator_i</i>	
	C sector	S sector
Intercept	1.170 (0.020)	1.122 (0.012)
ℓ_i	-0.212 (0.017)	-0.216 (0.009)
Observations	1,811	10,532

Note: Data source: 2003 ECINF. Standard errors in parentheses.

TABLE IV
FIRM-LEVEL LOG REVENUE PER WORKER AND LOG WAGES VS. LOG EMPLOYMENT.

Sector / Firm Type	A. Dep. Variable: $\log(\text{Revenue}_i/\ell_i)$				B. Dep. Variable: $\log(\text{wage}_i)$			
	Cf	Sf	Ci	Si	Cf	Sf	Ci	Si
Intercept	10.118 (0.013)	10.004 (0.005)	8.356 (0.044)	8.772 (0.019)	8.515 (0.002)	8.495 (0.001)	7.928 (0.039)	8.309 (0.018)
$\log(\ell_i)$	0.000 (0.005)	-0.128 (0.003)	0.453 (0.102)	0.401 (0.049)	0.110 (0.001)	0.123 (0.000)	0.334 (0.085)	0.293 (0.046)
<i>Exporter_i</i> (Indicator)	1.462 (0.021)				0.473 (0.006)			
Observations	16,986	43,861	1,608	8,858	140,649	949,995	1,609	8,861
Dataset	PIA + SECEX	PAS + PAC	ECINF	ECINF	RAIS + SECEX	RAIS	ECINF	ECINF

Note: Standard errors in parentheses.

TABLE V
TURNOVER, FIRM SIZE AND EXPORT STATUS.

	Dep. Variable: <i>Turnover_i</i>	
Intercept	0.713 (0.003)	0.633 (0.001)
$\log(\ell_i)$	-0.128 (0.001)	-0.116 (0.001)
<i>Exporter_i</i> (Indicator)	0.084 (0.007)	
Observations	140,649	949,995

Note: Data Sources: 2003 and 2004 RAIS and 2003 SECEX. Turnover of firm i between 2003 and 2004 is measured as $\text{Turnover}_i = \frac{|\ell_{i,2004} - \ell_{i,2003}|}{0.5 \times (\ell_{i,2004} + \ell_{i,2003})}$. Standard errors in parentheses.

- Table IV also shows a sizable formal wage premium, especially for small firms. In manufacturing the premium is about 0.59 log points at size one and 0.23 at size five.
- These tables together justify the core model assumption: informal firms are smaller, lower paying, and less productive, but they survive because they face fewer effective distortions.

5.4 Table V and Figures 1–2: Turnover patterns and the model's timing

Read:

- Table V shows that turnover declines strongly with firm size: the coefficient on log employment is about -0.128 in manufacturing and -0.116 in services.
- In manufacturing, exporters have higher turnover conditional on size, with an exporter coefficient of about 0.084.
- Figure 1 clarifies the incumbent timing: informal firms can remain informal, formalize, or exit; then they adjust labor, produce, and draw future productivity.
- Figure 2 clarifies the entrant timing: new firms pay entry costs, draw productivity, choose formal or informal status, and then choose labor. These diagrams help because the model is dynamic and the sorting problem is otherwise easy to lose track of.

5.5 Table VI: Fixed parameters

Read:

- The institutional environment is quantitatively heavy: payroll tax $\tau_w = 0.375$, value-added tax $\tau_y = 0.293$, import tariff 0.12, firing cost $\kappa = 1956.7$ reais, and minimum wage 2880 reais.

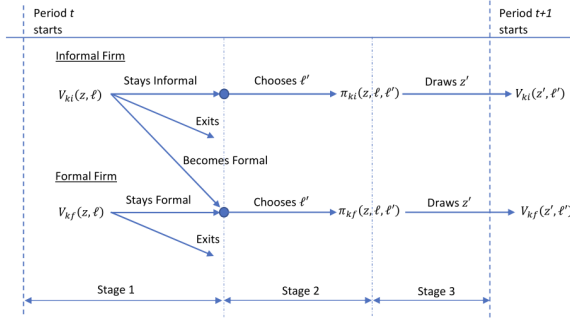


FIGURE 1.—Diagram of Firms' Behavior.

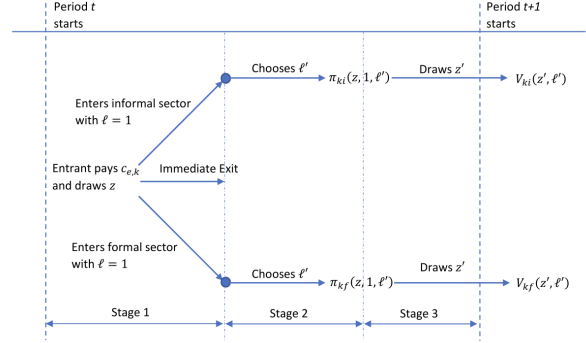


FIGURE 2.—Diagram of Entry Behavior.

- Input-output linkages are also strong: manufacturing's intermediate bundle puts weight $\lambda_C = 0.645$ on manufacturing, while services put weight $\lambda_S = 0.291$ on manufacturing.
- The benchmark iceberg trade cost is high at $\tau_c = 2.4$, which is why the counterfactual reductions in trade costs imply meaningful increases in openness.
- This table tells you that the large quantitative effects later in the paper do not come from an artificially frictionless baseline; they come from an already highly distorted economy.

TABLE VI
FIXED PARAMETERS.

Parameter	Description	Source	Value
τ_c	Iceberg Trade Cost	Head and Ries Index + 2003 WIOD	2.4
ξ	Share of final expend. on C	IBGE National Accounts (2000/2005)	0.296
λ_C	Prod. Function (Equation (4))	IBGE National Accounts (2000/2005)	0.645
λ_S	Prod. Function (Equation (4))	IBGE National Accounts (2000/2005)	0.291
r	Interest rate	Ulyssea (2010)	0.08
τ_y	Value Added Tax	Ulyssea (2018)	0.293
τ_w	Payroll Tax	Ulyssea (2018)	0.375
$\tau_u - 1$	Import Tariff	UNCTAD TRAINS	0.12
κ	Firing Costs (in R\$)	Heckman and Pagés (2000)	1,956.7
w	Min. Wage (in R\$)	Annualized 2003 value	2,880
b_u	Unemployment Benefit	$1.37 \times 5 = 6.85$ monthly Min. Wage	1,644
ξ	Matching Function	Petrongolo and Pissarides (2001)	0.5
ϕ	Matching Function	Own Calculations, see Section 4.3	0.576
β	Workers' Bargaining Weight	Mortensen and Pissarides (1999)	0.5

Note: Parameters based on the IBGE National Accounts employ simple averages between 2000 and 2005. WIOD stands for "World Input-Output Database."

TABLE VII
PARAMETER ESTIMATES.

Parameter	Description	$k = C$	$k = S$
$\tilde{a}_k$	Cost of Informality, Intercept	0.205	0.379
$\tilde{b}_k$	Cost of Informality, Convexity	0.071	0.010
$\tilde{h}_k$	Hiring Cost, Level	674.6	1,473.2
$\tilde{\gamma}_k^1$	Hiring Cost, Convexity	1.665	5.791
$\tilde{\gamma}_k^2$	Hiring Cost, Scale Economies	0.060	0.131
σ_k	Elasticity of Substitution	5.023	2.955
ρ_k	Productivity AR(1) Process, Persistence Coeff.	0.977	0.963
σ_k^2	Productivity AR(1) Process, Std. Dev. of Shock	0.210	0.409
α_k	Exogenous Exit Probability	0.069	0.081
$\bar{c}_k$	Fixed Cost of Operation	177.817	202.040
δ_k	Labor Share in Production	0.256	0.516
c_k^e	Entry Cost	4,616.5	4,893.8
f_k	Fixed Cost of Exporting		54,988.5
b_0	Utility Flow of Unemployment		-0.801
$(D_F^*)^{\frac{1}{\gamma_C}}$	Foreign Demand Shifter		1,847.4

Note: To aid interpretation, b_0 is expressed as a fraction of real income per capita according to the model.

5.6 Table VII: Estimated structural parameters

Read:

- Informality costs rise more steeply with size in manufacturing than in services, which fits the idea that larger manufacturing firms are harder to hide from the state.
- Hiring costs are convex in both sectors, especially in services, which helps the model generate wage dispersion and realistic turnover behavior.
- Elasticities of substitution are in a reasonable range: roughly 5 in manufacturing and 3 in services.
- Productivity is very persistent, and services are subject to larger firm-level shocks than manufacturing.

TABLE VIII
IMPACTS OF A SMALL TRADE COST SHOCK, τ_c DECLINES FROM 2.4 TO 2.3.

	Real Income	Welfare
1. $\Delta \log \mathcal{Y}$	1.317	1.375
2. $\Delta \log \mathcal{Y}^{\text{ME}}$	0.422	0.493
3. Δ Reallocation Effect (37)	0.895	0.882
4. $\Delta \log \mathcal{Y}^{\text{DF}}$	0.580	0.678
5. Δ Reallocation Effect (39)	0.737	0.698
6. Δ Share Unemployment	0.180	
	C Sector	S Sector
7. Δ Share Informality _k	-1.142	-0.569
8. $\Delta \log TFP_k$	1.988	1.042
9. $\Delta \log \text{Var}(\log(\partial R_k(z, \ell)/\partial \ell))$	-1.105	-1.291

Note: All numbers are expressed as $100 \times$ changes relative to the benchmark with $\tau_c = 2.4$. Total welfare is given by $\mathcal{Y} = \text{Real Income} + b_d L_u$, where L_u is the mass of unemployed workers. Δ Reallocation Effect (37) is obtained using equation (37), and Δ Reallocation Effect (39) using equation (39). Share Informality_k is the share of sector $k \in \{C, S\}$ employment in informal firms. TFP_k is aggregate total factor productivity in sector k , defined as the employment-weighted average of firm-level productivities z .

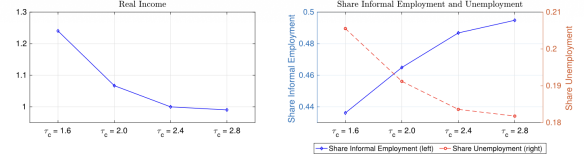


FIGURE 3.—Real Income, Informality, and Unemployment. Note: Real income in the left panel is displayed relative to its benchmark value under $\tau_c = 2.4$.

5.7 Table VIII and Figure 3: Small and large trade-cost reductions

Read:

- For a small fall in trade costs from $\tau_c = 2.4$ to 2.3, real income rises by 1.317%. The mechanical effect is only 0.422%, while the reallocation effect under decomposition (37) is 0.895%.
- Under the distortion-free decomposition, the trade gain in an undistorted economy is 0.580%, whereas the benchmark economy gets an additional 0.737% from reallocation. So the full gain is about 2.3 times as large as the distortion-free gain.
- The same small trade shock reduces informal employment by about 1.142 percentage points in manufacturing and 0.569 in services, while raising measured TFP by about 1.988% and 1.042%, respectively.
- Figure 3 shows that for bigger reforms the gains become very large: lowering τ_c from 2.4 to 2.0 raises real income by about 6.7%, and lowering it to 1.6 raises real income by about 24%. Yet the total informal-employment share falls only modestly, from 48.7% to 43.6%.

5.8 Table IX, Table X, and Figure 4: Enforcement scenarios and why informality amplifies trade gains

TABLE IX
EFFECTS OF INCREASING THE COST OF INFORMALITY.

	Bench	Strict Enf. 1	Strict Enf. 2	Strict Enf. 3	No Inf.
1. Share Informal Emp.	0.487	0.403	0.335	0.267	0.000
2. Unemployment Rate	0.184	0.193	0.202	0.214	0.278
3. $s = \tau_c \text{Imports} / X_C$	0.117	0.117	0.124	0.129	0.143

TABLE X
IMPACTS OF A SMALL TRADE SHOCK ACROSS ENFORCEMENT SCENARIOS, τ_c DECLINES FROM 2.4 TO 2.3—ALL IMPACTS SHOWN ARE RELATIVE TO THOSE UNDER THE BENCHMARK.

	Bench	SE1	SE2	SE3	No Inf.
1. $\Delta \log(\text{Real Income})$	0	-0.083	-0.856	-0.943	-0.500
2. $\Delta \log(\text{Real Income})^{\text{DF}}$	0	0.005	0.044	0.074	0.169
3. Δ Reallocation Effect (39)	0	-0.088	-0.900	-1.017	-0.669

Note: All effects are relative to the Benchmark and multiplied by 100. More precisely, each cell is given by $100 \times (\Delta)_{\text{scenario}} - \Delta_{\text{Bench}}$, where $y \in (\log(\text{Real Income}), \log(\text{Real Income})^{\text{DF}}, \text{Reallocation Effect})$ and scenario $\in (\text{Bench}, \text{SE1}, \text{SE2}, \text{SE3}, \text{No Inf.})$. Δ Reallocation Effect (39) is the reallocation effect obtained using equation (39).

Read:

- Table IX shows what stricter enforcement does before any new trade shock: the informal employment share falls from 48.7% in the benchmark to 40.3%, 33.5%, 26.7%, and then 0 in the no-informality scenario.
- But stricter enforcement also raises unemployment, from 18.4% in the benchmark to 27.8% with no informality, largely because the minimum wage becomes more binding when firms can no longer evade it informally.

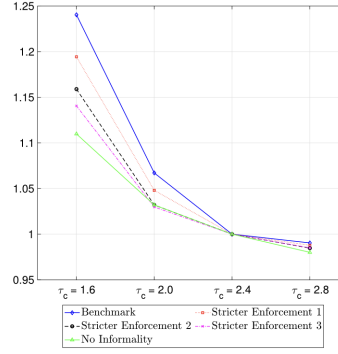


FIGURE 4.—Effect of Trade on Real Income Relative to $\tau_c = 2.4$, Various Enforcement Scenarios.

- Table X shows that the real-income gains from the small trade shock are smaller in every stricter-enforcement scenario than in the benchmark. The reason is not the mechanical effect; it is the smaller reallocation effect.
- Figure 4 delivers the big-picture comparative static: with the large liberalization from $\tau_c = 2.4$ to 1.6, the benchmark economy gains 24%, the mildly stricter-enforcement economy gains 19.4%, and the no-informality economy gains only 11%.

5.9 Table XI, Table XII, and Figure 5: Negative labor-demand shocks and the “employment buffer” role of informality

TABLE XI

IMPACT OF A NEGATIVE PRODUCTIVITY SHOCK OF 1 PCT—BENCHMARK SCENARIO.

	Benchmark
1. Δ Share Workforce Informal	5.377
2. Δ Share Unemployment	−1.058
3. $\Delta \log(\text{Real Income})$	−10.435
4. $\Delta \log(\text{Real Income})^{\text{DF}}$	−2.685
5. Δ Reallocation Effect (39)	−7.750

Note: All numbers are expressed in 100× changes relative to the initial equilibrium. Δ Reallocation Effect (39) is the reallocation effect obtained using equation (39).

TABLE XII

IMPACTS OF A NEGATIVE PRODUCTIVITY SHOCK OF 1 PCT ACROSS ENFORCEMENT SCENARIOS—ALL IMPACTS ARE RELATIVE TO THOSE UNDER THE BENCHMARK.

	Bench	SE1	SE2	SE3	No Inf.
1. Δ Share Workforce Informal	0	−1.883	−2.291	−3.350	−5.377
2. Δ Share Unemployment	0	0.461	0.642	0.970	2.517
3. $\Delta \log(\text{Real Income})$	0	3.852	4.882	5.903	5.198
4. $\Delta \log(\text{Real Income})^{\text{DF}}$	0	−0.002	−0.019	−0.032	−0.073
5. Δ Reallocation Effect (39)	0	3.855	4.901	5.935	5.270

Note: All effects are relative to the Benchmark and multiplied by 100. More precisely, each cell is given by $100 \times (\Delta)_{\text{scenario}} - \Delta)_{\text{Bench}}$. $y \in (\log(\text{Real Income}), \log(\text{Real Income})^{\text{DF}}, \text{Reallocation Effect})$, scenario $\in \{\text{Bench}, \text{SE1}, \text{SE2}, \text{SE3}, \text{No Inf.}\}$. Δ Reallocation Effect (39) is the reallocation effect obtained using equation (39).

Read:

- Table XI studies a 1% negative productivity shock in the benchmark economy. Informal employment rises by 5.377 percentage points, unemployment actually falls slightly by 1.058 percentage points, and real income falls by 10.435%.
- The key message is that informality absorbs displaced workers, consistent with the reduced-form evidence, but it does so by reallocating resources toward less distorted and less productive firms.
- Table XII shows that when enforcement is stricter, the increase in informality is smaller and the increase in unemployment is larger. Yet the real-income decline is *less* severe than in the benchmark because the negative reallocation effect is smaller.
- Figure 5 visualizes the productivity side: adverse shocks lower aggregate TFP more in the benchmark economy than in stricter-enforcement economies because the shock reallocates activity toward the informal sector.

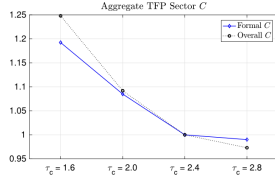


FIGURE 5.—Trade and Aggregate Productivity, Relative to Benchmark ($\tau_c = 2.4$).

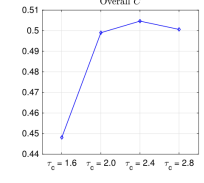
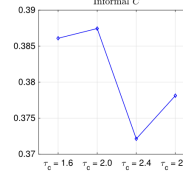
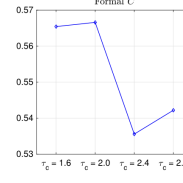
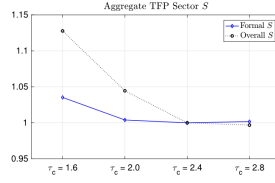


FIGURE 6.—Trade and Wage Inequality in Sector C, Standard Deviation of log Wages Across Workers.

5.10 Figure 6: Trade and wage inequality

Read:

- The two left panels show that trade liberalization increases wage inequality across firms within formal manufacturing and within informal manufacturing.
- That result lines up with earlier work focused on the formal sector only.
- But the right panel shows that once between-sector wage differences are included, overall wage inequality across workers in manufacturing can fall with trade openness.
- So the paper's message is not merely that earlier results are wrong; it is that they are partial-equilibrium or partial-sample results that need to be reinterpreted once informality is included.

6 Short summary

- **Conclusion.** Trade liberalization in a high-informality economy does much more than lower import prices. It also reallocates labor and output away from less distorted but low-productivity informal firms toward more distorted but high-productivity formal firms.
- **Identification insight.** The model is disciplined by a broad collection of moments on employment, transitions, size, wages, turnover, and exports across both formal and informal sectors.
- **Main empirical lesson.** Informality amplifies the gains from trade by creating larger ex ante misallocation, but it also cushions unemployment under negative labor-demand shocks while worsening the aggregate efficiency consequences of those shocks.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that the interaction between trade and domestic distortions can be quantitatively first-order in developing countries.
- It shows that when taxes and labor regulations are imperfectly enforced, formal and informal firms are not simply two sectors of the economy; they are two distortion regimes.
- In that environment, the reallocation effect of trade can dominate the usual mechanical effect.
- The magnitude of the gains from trade depends on how distorted the initial allocation is, which in this paper is tightly linked to the prevalence of informality.

7.2 Economic intuition

- Informality protects small, low-productivity firms from taxes and regulation. That makes them privately viable but socially too large.
- Formality exposes larger, more productive firms to taxes, payroll costs, firing costs, and minimum wages. That makes them socially too small.
- Trade liberalization shifts demand, profitability, and labor allocation in a way that favors the more productive formal side of the economy, so it partially undoes the domestic misallocation created by imperfect enforcement.
- But the same informal margin that helps displaced workers find employment during bad shocks can also redirect resources toward less productive uses, worsening welfare in downturns.

7.3 Takeaway

This paper is best thought of as a bridge between the trade, informality, misallocation, and labor-search literatures. Its central message is that the informal sector changes both the *size* and the *interpretation* of trade effects. In a distorted developing economy, trade can generate unusually large gains because it reallocates resources toward firms that are initially too constrained by formal regulations. At the same time, informality changes how the economy responds to negative shocks: it softens unemployment responses but can intensify the real-income cost by amplifying misallocation. For work on trade reform, labor markets, tax enforcement, or informality in developing countries, this paper is a benchmark reference.